UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 13

Fall 2025

Instructor: Somayeh Sojoudi

Typeset by: George Ma

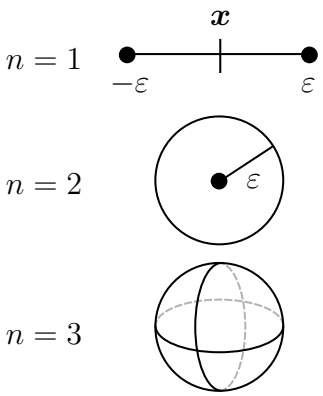
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Convexity.
First, we define different types of sets.

- A ball in $\mathbb{R}^n$ centered at $\boldsymbol{x} \in \mathbb{R}^n$ of radius ε is shown as

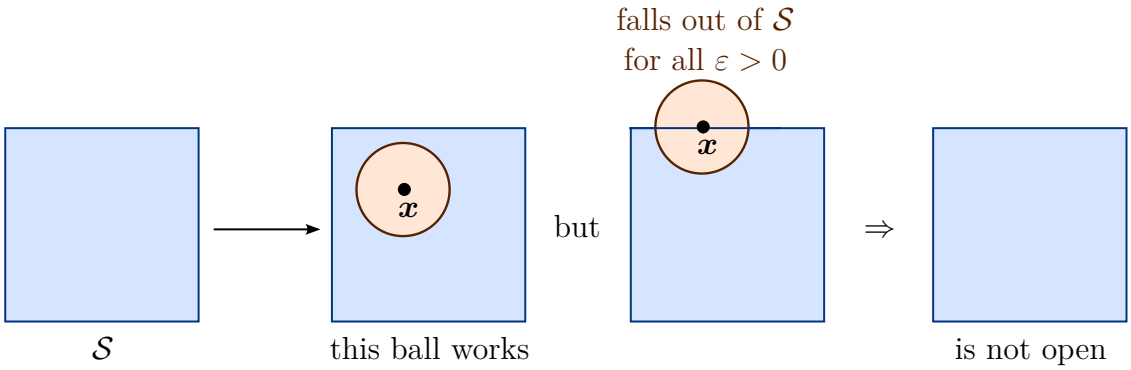
$$B_\varepsilon(\boldsymbol{x}) = \{\boldsymbol{z} \in \mathbb{R}^n \mid \|\boldsymbol{z} - \boldsymbol{x}\|_2 \leq \varepsilon\}.$$

  Example:

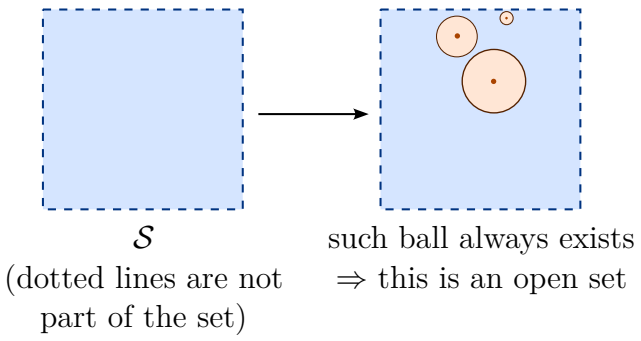


- A set $\mathcal{S} \in \mathbb{R}^n$ is called *open* if for every $\boldsymbol{x} \in \mathcal{S}$, there is a radius $\varepsilon > 0$ such that $B_\varepsilon(\boldsymbol{x}) \subseteq \mathcal{S}$.

  Example:



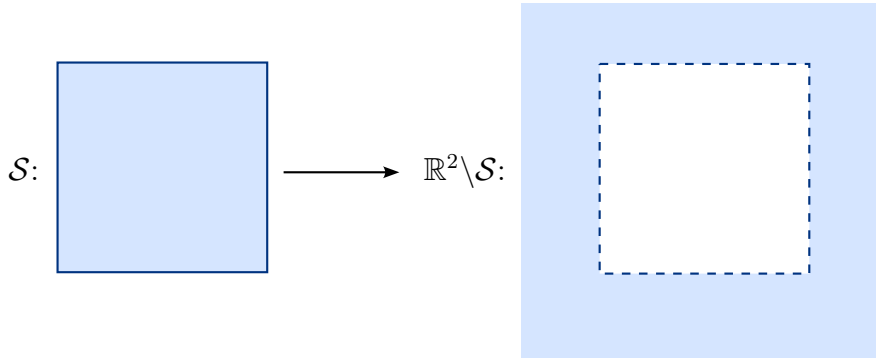
  The above set is not open because if $\boldsymbol{x}$ is on the edge of $\mathcal{S}$, some part of the ball $B_\varepsilon(\boldsymbol{x})$ always falls out of $\mathcal{S}$ no matter how you choose $\varepsilon > 0$. However, the following set is open:



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Notice that the edges are no longer part of $\mathcal{S}$.

- A set $\mathcal{S} \subseteq \mathbb{R}^n$ is called *closed* if its complement $\mathbb{R}^n \backslash \mathcal{S}$ is open.
  Example:



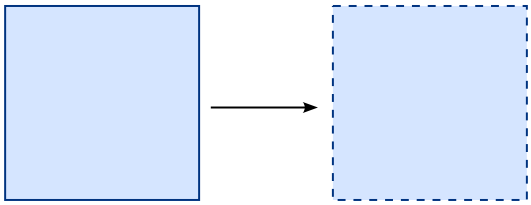
  $\mathbb{R}^2\backslash\mathcal{S}$ is open, thus $\mathcal{S}$ is closed.

- $\mathbb{R}^n$ and the empty set $\varnothing$ are both open and closed.

- The *interior* of a set $\mathcal{S} \subseteq \mathbb{R}^n$ is defined as:

$$\text{int}(\mathcal{S}) = \{\boldsymbol{z} \in \mathcal{S} \mid B_\varepsilon(\boldsymbol{z}) \subseteq \mathcal{S} \text{ for some } \varepsilon > 0\}.$$

  The interior includes all points $\boldsymbol{z}$ that are strictly inside the set, meaning that there is a ball around each such point that is in the set:



dotted lines are
excluded from the set

Example 1 (Interior of sets)**.**

$$\text{int}(\mathbb{R}^n) = \mathbb{R}^n, \quad \text{int}(\underbrace{[1,2]}_{1\le x\le 2}) = \underbrace{(1,2)}_{1<x<2}.$$

■

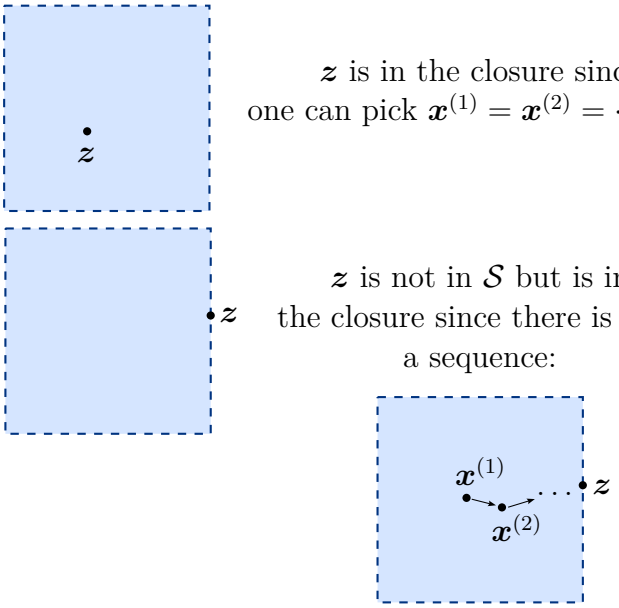
- The *closure* of a set $\mathcal{S} \subseteq \mathbb{R}^n$ is defined as:

$$\text{cls}(\mathcal{S}) = \left\{\boldsymbol{z} \in \mathbb{R}^n \;\middle|\; \boldsymbol{z} = \lim_{k\to\infty} \boldsymbol{x}^{(k)} \text{ where } \boldsymbol{x}^{(k)} \in \mathcal{S},\ \forall k\right\}.$$

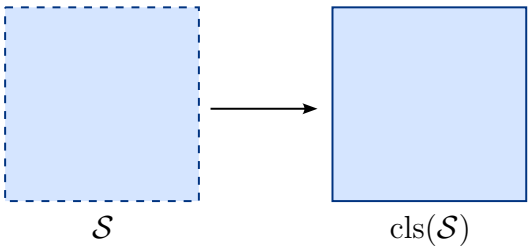
  This means that a point $\boldsymbol{z}$ is in the closure if there is a sequence of points $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots$ in $\mathcal{S}$ that converges to $\boldsymbol{z}$.

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Example:



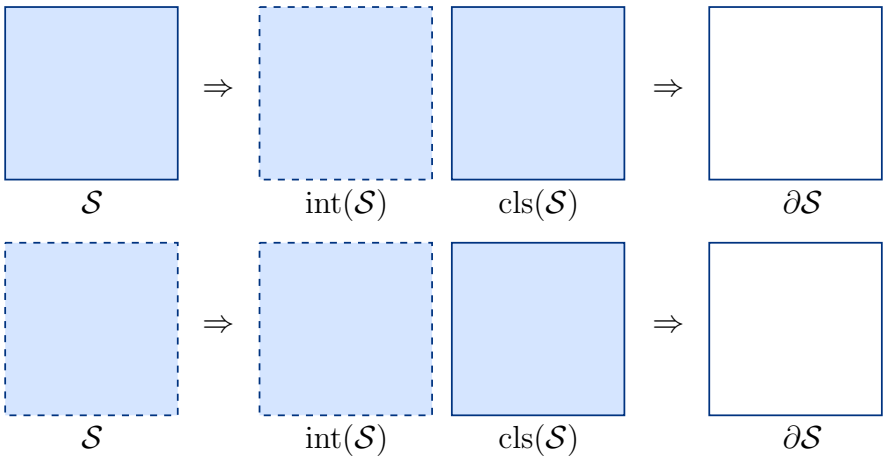
The closure includes points on the boundary:



- The *boundary* of a set $\mathcal{S} \subseteq \mathbb{R}^n$ is defined as

$$\partial\mathcal{S} = \text{cls}(\mathcal{S}) \backslash \text{int}(\mathcal{S}).$$

Example:

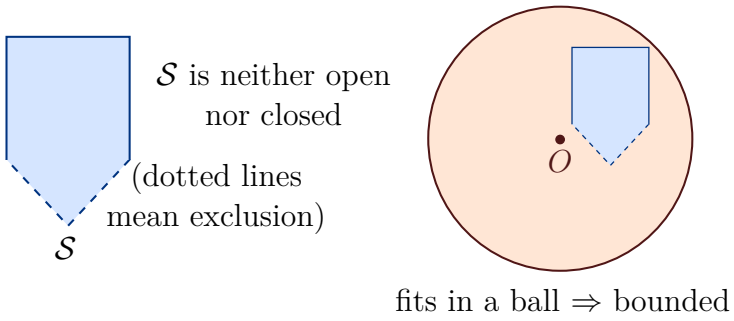


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Note that in the second case, $\partial\mathcal{S}$ is not part of $\mathcal{S}$.

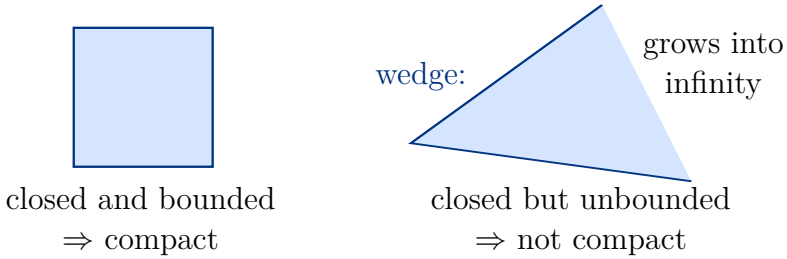
- A set $\mathcal{S} \subseteq \mathbb{R}^n$ is *bounded* if there exists $r > 0$ such that $\mathcal{S} \subseteq B_r(\mathbf{0})$ (meaning that $\mathcal{S}$ can be fit into a ball centered at the origin).

  Example:



- A set $\mathcal{S} \subseteq \mathbb{R}^n$ is called *compact* if it is closed and bounded.

  Example:



Consider a set of points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)} \in \mathbb{R}^n$.

- A point $\boldsymbol{x} \in \mathbb{R}^n$ is called a *linear combination* of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ if $\boldsymbol{x} = \alpha_1\boldsymbol{x}^{(1)} + \cdots + \alpha_m\boldsymbol{x}^{(m)}$ for some $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$.
- The set of all linear combinations of points is called the *linear hull* of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$, which is the same as $\text{span}(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)})$.
- A point $\boldsymbol{x} \in \mathbb{R}^n$ is called an *affine combination* of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ if $\boldsymbol{x} = \alpha_1\boldsymbol{x}^{(1)} + \cdots + \alpha_m\boldsymbol{x}^{(m)}$, for some $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$ such that $\alpha_1 + \cdots + \alpha_m = 1$.
- A point $\boldsymbol{x} \in \mathbb{R}^n$ is called a *convex combination* of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ if $\boldsymbol{x} = \alpha_1\boldsymbol{x}^{(1)} + \cdots + \alpha_m\boldsymbol{x}^{(m)}$, for some $\alpha_1, \ldots, \alpha_m \in \mathbb{R}$ such that $\alpha_1, \ldots, \alpha_m \geq 0$ and $\alpha_1 + \cdots + \alpha_m = 1$.

Definition 1 (Affine hull)**.** Given a set $\mathcal{S} \subseteq \mathbb{R}^n$, define the *affine hull* of $\mathcal{S}$ as the set of all affine combinations of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)} \in \mathcal{S}$ for all possible points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ and all values of $m \in \{1, 2, \ldots\}$.

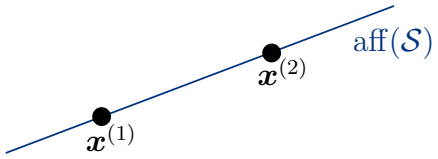
Theorem 1. *The affine hull* $\text{aff}(\mathcal{S})$ *of an arbitrary set* $\mathcal{S}$ *is an affine set.*

(Recall that an affine set is defined to be a point plus a subspace in Definition 4 of Lecture Note 2.)

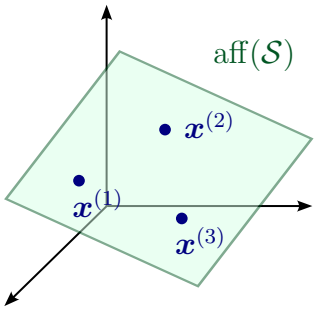
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Example 2. 1. If $\mathcal{S} = \{\boldsymbol{x}^{(1)}\}$, then $\text{aff}(\mathcal{S}) = \{\boldsymbol{x}^{(1)}\}$.

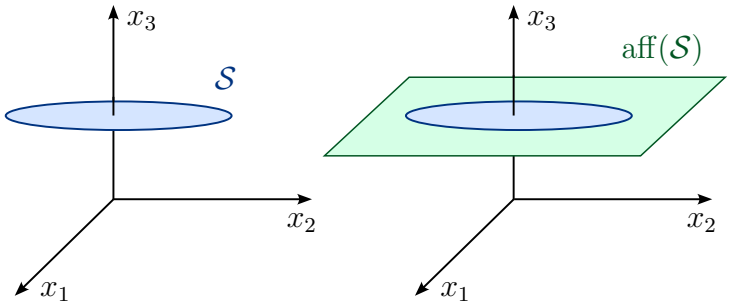
2. If $\mathcal{S} = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}\}$, then $\text{aff}(\mathcal{S})$ is a line passing through $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$.



3. If $\mathcal{S} = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)}\}$, then $\text{aff}(\mathcal{S})$ is a plane passing through $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \boldsymbol{x}^{(3)}$.



4. If $\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^3 \mid x_3 = 1,\ x_1^2 + x_2^2 \leq 1\}$, then $\mathcal{S}$ is a lifted ball, and $\text{aff}(\mathcal{S})$ is a plane containing this lifted ball.



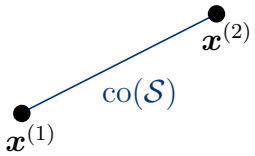
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Definition 2 (Convex hull)**.** Given a set $\mathcal{S} \subseteq \mathbb{R}^n$, define the *convex hull* of $\mathcal{S}$ as the set of all convex combinations of $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)} \in \mathcal{S}$ for all possible points $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}$ and all values of $m \in \{1, 2, \ldots\}$.

We denote the convex hull of $\mathcal{S}$ as $\text{co}(\mathcal{S})$.

Example 3. 1. If $\mathcal{S} = \{\boldsymbol{x}^{(1)}\}$, then $\text{co}(\mathcal{S}) = \{\boldsymbol{x}^{(1)}\}$.

2. If $\mathcal{S} = \{\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}\}$, then $\text{co}(\mathcal{S})$ is the line segment connecting $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$.



3. If $\mathcal{S} = \{\boldsymbol{x} \in \mathbb{R}^2 \mid x_1^2 + x_2^2 = 1\}$ is a circle, then $\text{co}(\mathcal{S}) = \{\boldsymbol{x} \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\}$ is a ball.

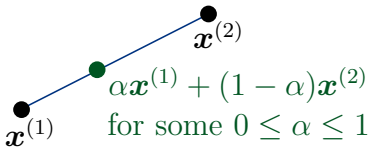
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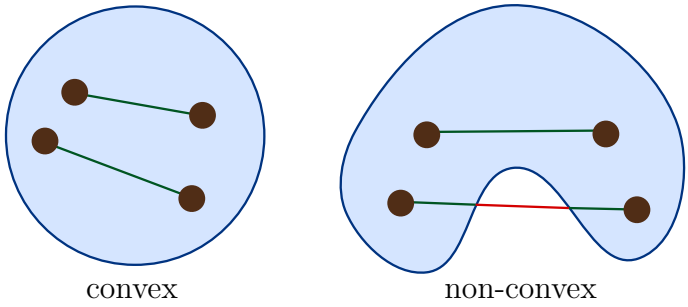
Definition 3 (Convex set)**.** A set $\mathcal{C} \subseteq \mathbb{R}^n$ is called *convex* if for every two points $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} \in \mathcal{C}$, the segment connecting $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$ is in $\mathcal{C}$.

In other words,

$$\underbrace{\alpha \boldsymbol{x}^{(1)} + (1-\alpha)\boldsymbol{x}^{(2)}}_{\text{characterization of segment}} \in \mathcal{C}, \qquad \forall \boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} \in \mathcal{C},\ \forall \alpha \in [0,1].$$



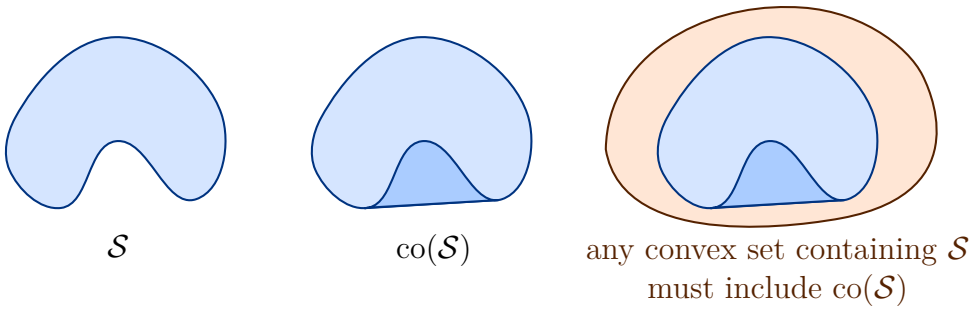
Example 4 (Convex and non-convex sets)**.**



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Theorem 2. *Given an arbitrary set $\mathcal{S} \subseteq \mathbb{R}^n$, its convex hull* $\text{co}(\mathcal{S})$ *is the smallest convex set containing $\mathcal{S}$, meaning that any other convex set containing $\mathcal{S}$ includes* $\text{co}(\mathcal{S})$.

Example:



(The convex hull adds the dark blue part to $\mathcal{S}$.)

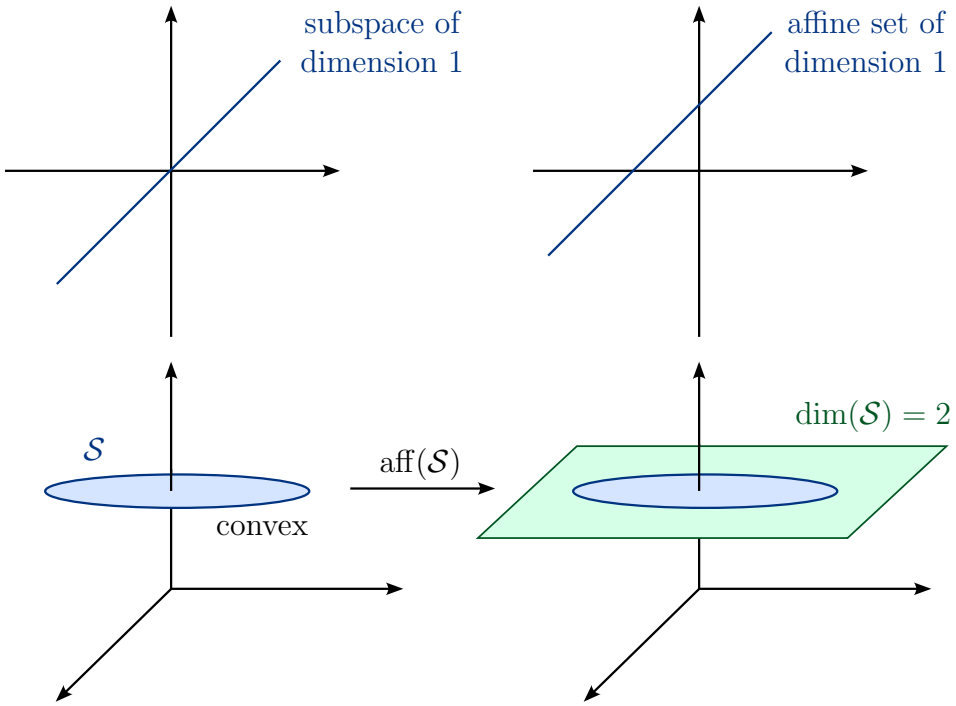
Definition 4 (Dimension)**.** Consider a set $\mathcal{S} \subseteq \mathbb{R}^n$.

1. If $\mathcal{S}$ is a subspace, then $\dim(\mathcal{S})$ is the number of basis vectors.
2. If $\mathcal{S}$ is an affine set: $\mathcal{S} = \boldsymbol{x}_0 + V$, where $\boldsymbol{x}_0 \in \mathbb{R}^n$ is a point and $V \subseteq \mathbb{R}^n$ is a subspace, then $\dim(\mathcal{S})$ is defined as $\dim(V)$.

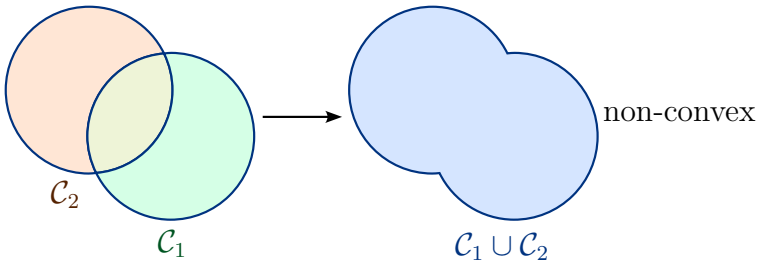
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3. If $\mathcal{S}$ is a convex set, then $\dim(\mathcal{S})$ is defined as the dimension of the affine hull $\text{aff}(\mathcal{S})$.

Examples:

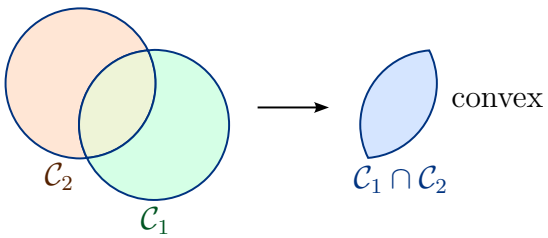


The union of two convex sets may not be convex:



Theorem 3. *Given convex sets $\mathcal{C}_1, \ldots, \mathcal{C}_m$, their intersection $\mathcal{C}_1 \cap \mathcal{C}_2 \cap \cdots \cap \mathcal{C}_m$ is convex.*

Example:



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